CSM
Fall 2026

EE 64
Worksheet 7

1. Phasor and Impedance Basics

(a) What is Euler's identity?

(b) Check that $\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$ holds using Euler's identity.

(c) What is the phasor form of the signal?

$$v(t) = V_0 \cos(\omega t + \phi_v) \tag{1}$$

(d) What is the magnitude and phase of the expression $8e^{j\frac{\pi}{3}}$?

(e) Transform the signal $12\cos\left(\omega t - \frac{3\pi}{4}\right)$ to phasor domain, in the form $Ae^{jB\pi}$.

(f) What is the impedance of a capacitor C?

(g) What is the impedance of an inductor L?

(h) What is the impedance of a resistor R?

(i) What is the impedance of an resistor R and a capacitor C in parallel?

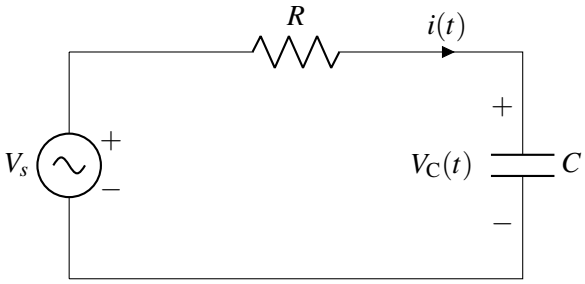
(j) What is the impedance of an inductor L and a capacitor C in parallel?

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2. Phasor Analysis

Equipped with the phasor representation, let's see how using it can greatly simplify alternating current (AC) circuit analysis, assuming that the voltages and currents vary with time as sinusoids.

The phasor analysis method consists of five steps. We'll walk through it in the RC circuit below:



The voltage source is given by

$$V_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right), \tag{2}$$

with $\omega = 10^3$ rad/s, $R = 1$ kΩ, and $C = 1$ μF.

Our goal is to obtain a solution for $V_C(t)$ and $i(t)$ with the sinusoidal voltage source $V_s(t)$.

(a) **Step 1: Convert $v(t)$ as a sum of complex exponentials.**
All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. **Convert $V_s(t)$ into an exponential and write down its phasor representation $\widetilde{V}_s$.**

(b) **Step 2: Transform circuits elements to phasor domain impedances.**
The voltage source is represented by its phasor $\widetilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\widetilde{I}$. **What are the phasor representations of R and C?**

(c) **Step 3: Cast KCL and/or KVL equations in phasor domain.**
Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.

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(d) **Step 4: Solve for unknown variables**

Solve the equation you derived in Step 3 for $\widetilde{I}$ and $\widetilde{V_C}$. What is the polar form of $\widetilde{I}$ ($Ae^{i\theta}$, where A is a positive real number)?

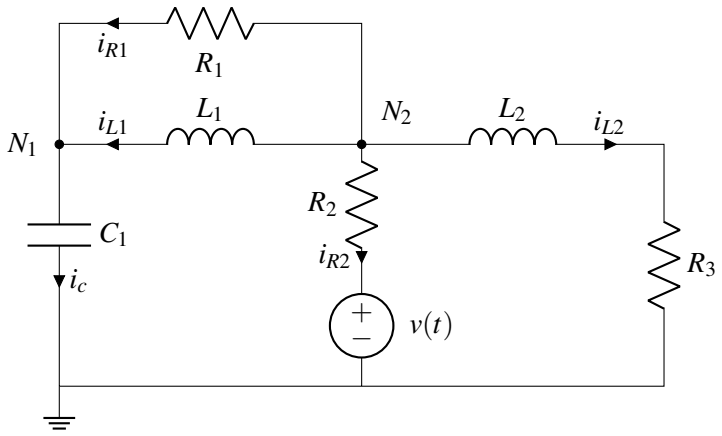
(e) **Step 5: Transform solutions back to time domain**

To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $V_C(t)$? What is the phase difference between $i(t)$ and $V_C(t)$?

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3. Phasor-domain circuit analysis

The analysis techniques you learned previously for resistive circuits are equally applicable for analyzing AC circuits (circuits driven by sinusoidal inputs) in the phasor domain. In this problem, we will walk you through the steps with a concrete example. Consider the circuit below.



The components in this circuit are given by:

Voltage source:

$$v(t) = 20\cos\left(50t - \frac{\pi}{3}\right)$$

Resistors:

$$R_1 = 8\Omega, \quad R_2 = 8\Omega, \quad R_3 = 8\Omega$$

Inductors:

$$L_1 = 40\text{ mH}, \quad L_2 = 40\text{ mH}$$

Capacitor:

$$C_1 = 5\text{ mF}$$

(a) To begin with, transform the given circuit to the phasor domain.

(b) Write out KCL for node N_1 and N_2 in the phasor domain.

(c) Use KVL to express the currents in terms of node voltages in the phasor domain. The node voltages $\widetilde{V}_1$ and $\widetilde{V}_2$ are the voltage drops from N_1 and N_2 to the ground.

(d) Write the equations you derived in part (b) and (c) in a matrix form, i.e., $A\begin{bmatrix}\widetilde{V}_1\\ \widetilde{V}_2\end{bmatrix} = b$ Solve for A and b in numerical form.

(e) Solve the systems of linear equations you derived in part (d) with any method you prefer, and then find $i_c(t)$.

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